

Wedges and welfare

June 2026

1 Introduction

This note purports to establish the following general result: One can do a full welfare analysis of a sector-specific policy reform (within a model with utility quasilinear in a numeraire good, i.e. the microfoundation required for classical economic surplus analysis) in two steps: 1) do a "naive" classical economic surplus analysis, i.e. adding up consumer surplus, producer surplus, and tax revenue, implicitly assuming that the wedge prevailing for the numeraire good is 0 2) do an "accounting adjustment" that can be done in either of the following ways: 2.1) for global welfare assessment: take the aggregate decrease in the value of inputs used for the sector under consideration and multiply this amount by the wedge for the numeraire good. This yields exactly the required addition to the economic surplus. 2.2) for disaggregated (by country) welfare assessment: Compute "change in tax revenue" (A), "change in producer surplus" (B) and "change in spending on numeraire, before taking into account that the taxes get transferred in a lump sum way to the consumer and that the producer surplus also accrue in a lump sum way to the consumer, given that he is the owner of the firms" (C) Multiply each with the wedge for the numeraire good.

Define $wedge_j$ to be the "wedge" prevailing in sector j , where this is defined via:

$$(1 + wedge_j)c_j = p_{consumer}c_j = (1 + t_j)(1 + m_j)c_j$$

Let us define "one util" to be the increase in well-being induced by one dollar of additional disposable income. By (auxiliary) assumption, we assume that this is the same across people. Suppose an additional unit (the amount that commands a price of 1 dollar) of the "numeraire resource" gets used for flights. The way that this gets converted into utils is at $1 + wedge_{aviation}$, given that this is the amount that the representative consumer would need to pay for a marginal unit of aviation services. This is already taken into account via the market-specific model.

Now the point where we need to do an adjustment to go from a market-specific analysis (that implicitly assumes that the wedge on the numeraire good is 0) is when it comes to evaluating the utility resulting from consumption of the numeraire good. In reality, the wedge is $wedge_{numeraire}$. Hence all that we need to do is to add $wedge_{numeraire} * "costs\ saved"$ as the "welfare component that we forgot". This accounting is easy if one tracks costs saved, but that's

difficult to combine with tracking to which country it accrues. Alternatively, one can do the accounting adjustment at the level of “change in tax revenue” (A), “change in producer surplus” (B) and “change in spending on numeraire, before taking into account that the taxes get transferred in a lump sum way to the consumer and that the producer surplus also accrue in a lump sum way to the consumer, given that he is the owner of the firms” (C) and compute the geometric series.

So for every dollar in (A+B+C) one has to take into account that in a first round, it translates into one util. But of that dollar, the proportion $\frac{wedge}{1+wedge}$ accrues as additional tax revenue or profits. These again benefit the consumer. An so on. So overall, we get per dollar in (A+B+C) the following series of utils:

$$1 + \frac{wedge}{1+wedge} + \left(\frac{wedge}{1+wedge}\right)^2 + \dots = \frac{1}{1 - \frac{wedge}{1+wedge}} = 1 + wedge$$

This provides a consistency check with the cost-based accounting defined above.

$$\begin{aligned} &v - p(1 + t) \\ &p - c \\ &tp \\ &v - c \end{aligned}$$